Girdhar Saini

AI/ML Engineer

Kaiserslautern, Germany | +(49)-15234735711 | https://www.linkedin.com/in/girdhar97 | girdharsaini.cvt@gmail.com | https://github.com/Girdhar97 | **Bilingual: German (B2)/English (C1)**

WORK EXPERIENCE

Forschungszentrum Informatik (FZI) | **AI-Research Assistant** *Karlsruhe, Germany | 05.2024 – 10.2025*

Project: EV Charging Machine Learning Forecasting Service for Energy Grid Management

- Architected and deployed production-level machine learning forecasting service on time-series data, with ETL data pipeline, data preprocessing and feature engineering
- Built REST API with Celery and Redis for distributed task queue processing; containerized with Docker to enable horizontal scalability and low-latency model predictions in milliseconds
- Implemented model monitoring using Prometheus and Grafana, tracking prediction latency, system performance metrics, and model drift in production
- Achieved 90%+ forecasting accuracy, reduced data noise by 34% and improved model reliability by 18%

Robotic Research Lab | **Master Thesis Student** *Kaiserslautern, Germany | 11.2024 - 05.2025*

Project: Optimizing Small Object Detection for Autonomous Vehicles in Dense Natural Environments

Grade: 1.3

- Architected modular PyTorch-based computer vision framework with 8+ phase-wise model configurations
- Improved real-time object detection mAP by +2.6% through enhancements to YOLOv5/YOLOv8, model optimization and hyperparameter tuning, leveraging HPC Linux clusters for distributed training
- Conducted model evaluation and performance benchmarking and inference speed analysis for real-time deployment feasibility

Continental Tires Deutschland GmbH, | **Data Science Intern** *Hannover, Germany | 08.2023 - 01.2024*

Project: Data-Driven Quality Control in Industrial Tire Apex Manufacturing

- Developed and deployed machine learning regression models for anomaly detection from time-series sensor data, improving defect identification by 25% through feature engineering
- Developed Flask and HTML/CSS-based defect detection software for process and quality control teams

EDUCATION

Technical University of Kaiserslautern-Landau | MSc Vehicle Technology *Kaiserslautern, Germany | 10.2021 - 10.2025*

- Specialization in Artificial Intelligence, final grade 1.9

Manipal University Jaipur | BTech Mechanical Engineering *Jaipur, India | 10.2015 - 10.2019*

TECHNICAL STACK

Programming: Python, C++, SQL

Machine Learning & AI: PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, NumPy, Pandas, Matplotlib

LLMs & Generative AI: Large Language Models (LLMs), LangChain, LangGraph, Hugging Face Transformers, Llama 3, OpenAI GPT-4 API, Groq API, Retrieval-Augmented Generation (RAG), Prompt Engineering, Fine-Tuning, Embedding Models, Vector Databases (FAISS, AstraDB)
MLOps & DevOps: Docker, Kubernetes, MLflow, GitHub Actions, CI/CD Pipelines, Jenkins, Model Deployment, Model Serving, Model Monitoring, Model Registry, Experiment Tracking, A/B Testing
Databases & Data Engineering: PostgreSQL, MongoDB, AstraDB, ETL Pipelines, Data Preprocessing, Data Transformation, Schema Validation
Cloud & Infrastructure: AWS (Lambda, EC2, S3), GCP (GKE, Cloud Run), HPC Platforms, Distributed Training
Monitoring & Observability: Prometheus, Grafana, Structured Logging, Latency Optimization, Model Drift Monitoring
Backend & APIs: FastAPI, Flask, REST APIs, Microservices Architecture, WebSockets
Frontend: React.js, Streamlit, HTML/CSS

PROJECTS

Project1: Flipkart-Style Product Recommender Chatbot *(Production LLMOps System)*

https://github.com/Girdhar97/Flipkart_Chatbot.git

- Built stateful e-commerce conversational AI delivering personalized product recommendations using Retrieval Augmented Generation (RAG) and semantic search
- Developed end-to-end data pipeline: CSV ingestion → HuggingFace BGE embeddings → AstraDB vector storage with metadata filtering for hybrid similarity search and low-latency retrieval
- Deployed scalable microservices architecture on GCP Kubernetes Engine with Docker multi-stage builds, Prometheus/Grafana monitoring stack tracking RAG latency, query throughput, and predictions/sec.

Project2: Stateful Agentic AI System with LangGraph & Llama 3 *(Multi-Agent LLM System)*

https://github.com/Girdhar97/AgenticChatbot.git

- Architected multi-agent Large Language Model (LLM) system with persistent conversation memory using LangGraph checkpointing and stateful workflow orchestration
- Built three specialized agents in unified architecture: conversational chatbot, tool-augmented coding assistant and AI news aggregation DAG with summarization capabilities using Llama 3 and Groq API

Project3: Network Security – Malicious URL Detection *(End-to-End MLOps Pipeline)*

machine_learning_project/Network_Security_ML_Project at main · Girdhar97/machine_learning_project

- Designed production-grade end-to-end ML pipeline for malicious URL classification with 94% F1-score using 30+ lexical and security features with feature engineering and model optimization
- Integrated MLflow on DagsHub for comprehensive experiment tracking, model registry, hyperparameter logging, and versioned model artifacts stored on AWS S3
- Engineered modular ETL workflow: MongoDB Atlas ingestion → YAML-driven schema validation → data preprocessing pipeline → automated model training and CI/CD deployment with GitHub Actions

Project4: ThyroCheck AI – Thyroid Cancer Detection with Explainable AI *(Computer Vision & XAI)*

https://github.com/Girdhar97/Computer_Vision_Projects/tree/main/Research_Based_Cancer_Detection_XAI

- Developed FibonacciNet CNN architecture in TensorFlow/Keras for medical image classification achieving over 90% accuracy and 0.96 AUC for thyroid cancer detection
- Integrated Grad-CAM explainable AI (XAI) generating class-discriminative attention heatmaps and automated clinical reports using python-docx
- Deployed FastAPI backend with dual frontend interfaces (Streamlit, React.js)